

SOM-GNN-SDM Switzerland

Species Distribution Modelling with Graph Neural Networks
at 30 m Resolution

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Motivation — Why Graphs for SDM?

Problem: Traditional SDMs (MaxEnt, RF) treat each location independently.
No spatial context — a pixel's prediction ignores its neighbours.

Insight: Habitat suitability is *not* independent.

A patch surrounded by suitable habitat is more valuable (connectivity, dispersal).

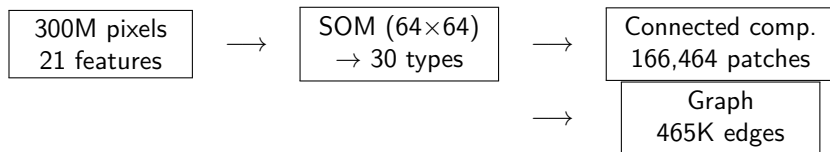
GNN-SDM (Wu et al. 2025): Encode the landscape as a graph.

- ▶ Nodes = habitat patches (with environmental features)
- ▶ Edges = spatial adjacency
- ▶ Message passing lets each node “see” its neighbourhood

This project: Adapt the framework to Swiss flora at 30 m resolution.

- ▶ Wu et al.: uniform $10'$ grid (~ 18.5 km cells), continental scale
- ▶ Ours: SOM-derived patches from 30 m pixels
- ▶ 3,756 plant species from GBIF, 21 environmental features

Pipeline — From Pixels to Graph

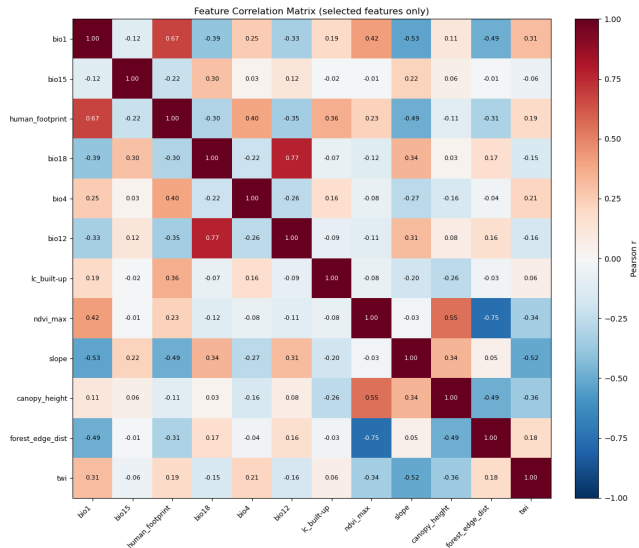


12 features per node (selected from 21 via RF importance):

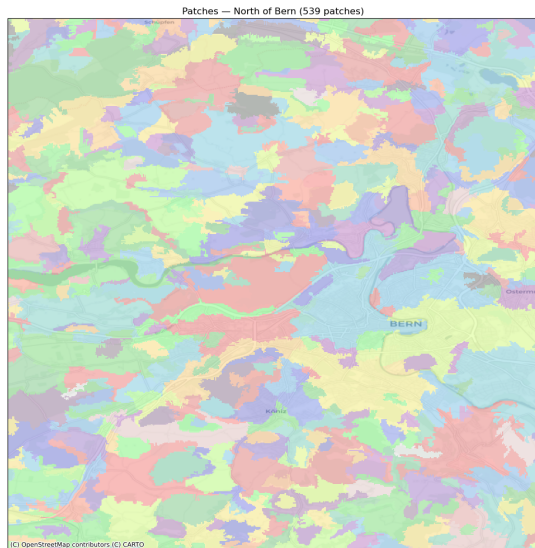
Group	Features	Source	Native res.
Terrain	elevation, slope, TWI	Copernicus GLO-30	30 m
Climate	bio1, bio4, bio12	CHELSAch 1981–2010	25 m
Land cover	tree, grass, built-up fractions	ESA WorldCover 2021	10 m
Biological	canopy height, NDVI, HFP	ETH / GIMMS / Mu et al.	10 m–1 km

All regridded to 30 m master grid, then mean-aggregated per patch → 12-dim node vector.

Feature Correlation Matrix



Landscape Patches — SOM-Based Segmentation



Model — GraphSAGE Architecture

```
class GNNSDM(torch.nn.Module):
    def __init__(self, in_channels, hidden_dims=[64, 48, 32], dropout=0.2):
        super().__init__()
        self.convs = torch.nn.ModuleList()
        prev = in_channels
        for dim in hidden_dims:
            self.convs.append(SAGEConv(prev, dim, aggr='mean'))
            prev = dim
        self.out = torch.nn.Linear(prev, 1)

    def forward(self, x, edge_index):
        for conv in self.convs:
            x = F.leaky_relu(conv(x, edge_index))
            x = F.dropout(x, p=self.dropout, training=self.training)
        return torch.sigmoid(self.out(x)).squeeze(-1)
```

Key design choices:

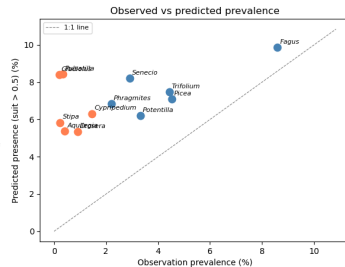
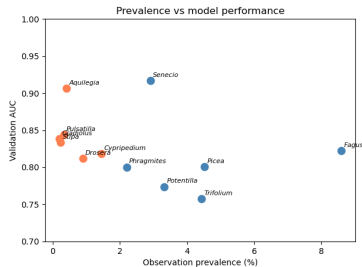
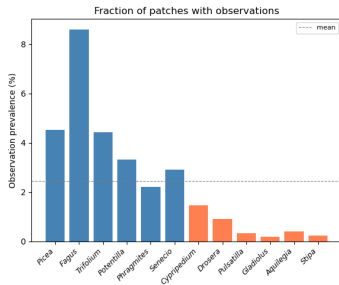
- ▶ Architecture search on 6 species → [64, 48, 32] beats paper's [24, 18, 8]
- ▶ Random pseudo-absences (3:1 ratio) — same as RF for fair comparison
- ▶ Node weighting via PageRank; weighted MSE loss; early stopping (patience=50)

Results — 12 Species Deep Dive (GNN vs RF)

Species	Group	Patches	GNN AUC	RF AUC	Δ
<i>Senecio inaequidens</i>	common	4,838	0.917	0.910	+0.007
<i>Fagus sylvatica</i>	common	14,305	0.822	0.788	+0.034
<i>Picea abies</i>	common	7,526	0.801	0.773	+0.028
<i>Phragmites australis</i>	common	3,662	0.800	0.787	+0.013
<i>Potentilla erecta</i>	common	5,527	0.774	0.760	+0.014
<i>Trifolium pratense</i>	common	7,383	0.757	0.720	+0.037
<i>Aquilegia alpina</i>	vulnerable	679	0.906	0.879	+0.027
<i>Pulsatilla vulgaris</i>	vulnerable	550	0.844	0.851	-0.007
<i>Gladiolus palustris</i>	vulnerable	329	0.838	0.838	≈ 0
<i>Stipa pennata</i>	vulnerable	392	0.834	0.844	-0.010
<i>Cypripedium calceolus</i>	vulnerable	2,423	0.818	0.797	+0.021
<i>Drosera rotundifolia</i>	vulnerable	1,499	0.812	0.812	≈ 0

- ▶ GNN consistently outperforms RF on common species (mean $\Delta = +0.022$)
- ▶ Vulnerable species: mixed — GNN wins where more presence patches exist
- ▶ Generalists (*Trifolium*) benefit most from graph context (+0.037)

Predicted vs Observed Prevalence



Next Steps — Conservation Application

What's done:

- ✓ Full pipeline: data → features → SOM → patches → graph → GNN
- ✓ 3,751 species with suitability scores
- ✓ RF baseline for comparison (same patches, no graph)
- ✓ IUCN threat status for all species

What's next:

1. **GNN vs RF full comparison** — quantify where graph structure helps
2. **Circuit-theory connectivity** — current flow through suitable habitat
3. **Gap analysis** — overlay with protected areas, find coverage gaps

```
# Circuit theory: resistance = 1 - suitability
# -> current flow identifies critical corridors
priority_score = (mean_suitability + mean_current_flow) / 2
```

Open questions:

- ▶ Does GNN help more for rare species? (connectivity matters more?)
- ▶ Can we lower the 100-record threshold for threatened species?